

How can I store both `StarPlayer` and `RegularPlayer` in one `ArrayList<Player>` and calculate their monthly salary using polymorphism instead of if-else statements in Java?

Thought for a few seconds >

Use a single list declared with the parent type:

```
Java
ArrayList<Player> players = new ArrayList<>();
```

Because both `StarPlayer` and `RegularPlayer` extend `Player`, they can both be stored in that list:

```
Java
players.add(new StarPlayer(
    "101", "Nguyen Van A", 22, "VNM",
    "Forward", 9, 1000, "active"
));

players.add(new RegularPlayer(
    "102", "Tran Van B", 24, "VNM",
    "Midfielder", 8, 800, "active"
));
```

In the abstract `Player` class, declare the salary method:

```
Java
public abstract double calculateMonthlySalary();
```

Then each subclass overrides it with its own rule:

```

// Java

public class StarPlayer extends Player {
    @Override
    public double calculateMonthlySalary() {
        return getBaseSalary() * 1.20;
    }
}

```

```

// Java

public class RegularPlayer extends Player {
    @Override
    public double calculateMonthlySalary() {
        return getBaseSalary() * 1.05;
    }
}

```

When displaying or creating salary records, loop through the list and call the same method:

```

// Java

for (Player player : players) {
    System.out.println("Player: " + player.getFullName());
    System.out.println("Monthly Salary: "
        + player.calculateMonthlySalary());
}

```

Java uses polymorphism to call the correct overridden method automatically. A `StarPlayer` receives the 20% calculation, while a `RegularPlayer` receives the 5% calculation, without needing `if-else` or `instanceof`.

```

public class PlayerList {
    private ArrayList<Player> players;

    private Player createPlayerByType(Scanner sc) {
        while (true) {
            System.out.print("Player Type (Star/Regular): ");
            String playerType = sc.nextLine().trim();

            if (playerType.equalsIgnoreCase("Star")) {
                return new StarPlayer();
            }
            if (playerType.equalsIgnoreCase("Regular")) {
                return new RegularPlayer();
            }
        }
    }

    // Dynamic dispatch: StarPlayer and RegularPlayer calculate salary differently.
    salary.setPlayerMonthlySalary(player.calculateMonthlySalary());
    salaryRecords.add(salary);
    System.out.println("Salary record added successfully.");
}

```

```
@Override
public double calculateMonthlySalary() {
    return getBaseSalary() * (1 + STAR_BONUS_RATE);
}
```

```
@Override
public double calculateMonthlySalary() {
    return getBaseSalary() * (1 + REGULAR_BONUS_RATE);
}
```